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CHENNAI

ARTIFICIAL INTELLIGENCE

BCSE306L

DIGITAL ASSIGNMENT – 1

COGNITIVE AGENTIC AI DIGITAL TWIN FOR SMART CITIES

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1. Problem Identification — Domain and Motivation

1.1 Application domain and why it matters

Rapid urbanization is concentrating an increasing share of the world's population in cities: roughly 55% of people live in urban areas today, and the United Nations projects this will rise to about 68% by 2050 [ref: UN World Urbanization Prospects, cited via WEF]. This concentration strains transportation networks, drainage and flood-control systems, power grids, and emergency-response capacity simultaneously, and the systems that manage these resources are largely siloed — traffic control, video surveillance, weather monitoring, and utility management run as separate platforms with no shared situational model.

The economic cost of failing to manage this complexity is large and measurable. The International Transport Forum estimates that traffic congestion alone costs the global economy on the order of US$1.4 trillion annually, and INRIX estimated US$87 billion in congestion-related losses for U.S. drivers in 2018 alone (about US$1,348 per driver). Urban flooding is a second, fast-growing risk: climate change and impervious urban surfaces are intensifying pluvial flood frequency, and even short-horizon flood-depth forecasting errors translate directly into failed evacuation and drainage-control decisions. A city administrator today must reconcile traffic cameras, IoT sensors, weather feeds, and GPS/fleet data manually or through disconnected dashboards, which is too slow for events (accidents, flash floods, infrastructure failures) that unfold over minutes.

Conventional “digital twins” in this space are largely descriptive: they visualize current sensor state on a 3D city model but do not predict, simulate counterfactual scenarios, or explain a recommended action. This project's premise is that a cognitive, agentic layer — combining computer vision, time-series forecasting, a shared urban knowledge graph, multi-agent reinforcement learning, and an LLM-based reasoning/explanation layer — can convert a passive monitoring twin into an autonomous decision-support platform.

1.2 Stakeholder and the decision supported

Primary stakeholder: municipal traffic-management and emergency-operations administrators (a City Operations Center role), who currently make manual, siloed decisions about signal timing, incident response dispatch, and flood/infrastructure alerts.

Decision supported: given the current and forecast state of city infrastructure (traffic, weather, incidents), the system recommends and explains a ranked set of interventions — e.g., re-time a set of signals, reroute a corridor, issue a flood alert for a district, or dispatch a maintenance/emergency crew — with an accompanying confidence score and rationale, rather than simply displaying raw sensor dashboards.

2. Literature Survey

The table below summarizes 18 references (≥ 15 required), of which 12 are from 2024–2026 (≥ 10 required) and are drawn from IEEE, Elsevier, Springer/Nature, MDPI Q1–Q2 journals, NeurIPS, and ACM venues. Three entries ([4], [9] in part, [15]) are preprints/under-review and are used only as supporting material, within the stated limit of three.

2.1 Comparison table

Ref.	Year	Dataset	Method / Architecture	Key Metric & Value	Stated Limitation
[1]	2025	150 QA-pair validation set (custom, urban)	Multi-agent LLM system + RAG over municipal info systems (Kalyuzhnaya et al.)	Pipeline-selection accuracy 94-99%; RAG improved response accuracy +17% (strategic) / +55% (service) ; G-Eval 0.68-0.74	Evaluated on a narrow 150-item QA benchmark; no live deployment or closed-loop control
[2]	2024	Custom China arterial network	Spatial-Temporal Graph Gated Transformer (STGGT) for traffic forecasting	Outperforms GCN/GRU/attention baselines on speed-forecast MAE/RMSE (city case study)	Single-city validation; no missing-modality or sensor-dropout analysis
[3]	2025	Urban arterial traffic dataset	Congestion-conditional adaptive spatiotemporal GNN with optimization	Reports improved congestion-prediction accuracy over GCN baselines	Optimization step adds inference latency; not tested under real-time constraints
[4]	2023	PeMS / synthetic auxiliary-info graphs	Sparse cross-attention GCN with auxiliary information awareness	Improves traffic-flow prediction MAE over GraphWaveNet/DCRNN baselines	Preprint; road-network auxiliary features not generalized to other cities
[5]	2024	Zhengzhou waterlogging records (15 points, 16 rainfall events)	Transformer-LSTM + Sparrow Search Algorithm hyper-parameter optimization	NSE 0.942, MAPE 30.24%, RMSE 0.049 m, MAE 0.038 m	Small event set (16 storms); single-city calibration limits transfer
[6]	2024	Shenzhen precipitation & flood-monitoring stations	Improved LSTM with integrated precipitation forecasting	$R^2 > 0.75$; RMSE reduced 13.8% at 8-step (40 min) horizon	Accuracy still degrades with longer lead times; single-city data
[7]	2024	USGS NHDPlus watershed dataset	Deep convolutional LSTM (ConvLSTM) with global attention for flash floods	Baseline LSTM RMSE = 0.07 ft; attention variant improves on elevated-flow cases	Performance still weak during extreme/elevated flood conditions
[8]	2026	Physics-based coupled-model flood database (3 flood-prone AOIs)	Coupled CNN + ConvLSTM spatiotemporal surrogate	NSE/RMSE/MAE reported per tile size; best at 100-cell tiling	Surrogate trained against a simulator, not ground-truth observed floods
[9]	2024	UCF-Crime, ShanghaiTech, MSAD, CUVA	10-year survey of deep video anomaly detection (CNN→transformer→VLM)	AUC improved from ~85% (early CNN) to >97% (transformer/VLM methods)	Cross-paradigm AUC comparability is questionable (weak- vs. unsupervised settings differ)
[10]	2023	UCSD Ped1/Ped2, Avenue, ShanghaiTech (multi-benchmark)	Background subtraction + convolutional autoencoder + object detector	Average frame-level AUC 94.94%; up to 7.7% AUC gain over prior SOTA	Autoencoder reconstruction struggles with crowded/occluded scenes
[11]	2024	SUMO 4x4 grid / real-world arterial network	Multi-agent DDQN for coordinated traffic-signal control	Waiting time −40-60%; queue length −40-60%; speed +10-18%	Gains are simulation-only; no field trial; reward design is not standardized
[12]	2024	SUMO signal-control benchmark (PN_D3QN)	Enhanced deep RL (dueling double DQN variant) for adaptive signal control	Faster convergence and lower average waiting time vs. baseline DQN/D3QN	Tested only in simulation; single intersection topology family
[13]	2026	5-domain Bangkok open datasets (transit, energy, environment, land use, demographics)	UrbanKG — layered knowledge-graph framework (287k triples) for cross-domain fusion	Cross-domain SPARQL query validation across all 5 domains	Proof-of-concept only; no production-scale governance or live-update evaluation
[14]	2024	OpenStreetMap, NYC OpenData, multi-city sources	UUKG — unified urban knowledge-graph dataset & benchmark for spatiotemporal prediction	Establishes benchmark tasks/metrics for KG-augmented urban prediction	Benchmark construction paper; does not itself propose a

Ref.	Year	Dataset	Method / Architecture	Key Metric & Value	Stated Limitation
					production decision-support agent
[15]	2024	Pirmasens city case study (accessibility & disaster data)	Dynamic knowledge-graph + agents for isochrone-based resilience/accessibility analysis	Demonstrates flood-time infrastructure-accessibility reasoning via critical-path analysis	Single mid-size city case study; agents are rule-driven, not learned
[16]	2024	Real building energy & BMS datasets (qualitative review)	Data-driven digital twins of buildings for smart-city energy modelling	Synthesizes implementation considerations; no single unified quantitative benchmark	Review paper; does not evaluate a working end-to-end digital twin
[17]	2025	Issy-les-Moulineaux case study	Modular Edge-Cloud digital-twin software architecture for smart cities	Demonstrates autonomous traffic-management viability in one deployment	Architecture-focused; limited quantitative accuracy reporting for the AI models used
[18]	2024	NeurIPS urban-KG construction benchmark	UrbanKGent — unified LLM-agent framework for automated urban knowledge-graph construction	Reduces manual KG-construction effort vs. rule-based pipelines (benchmark tasks)	Focus is KG construction, not downstream forecasting/decision-support integration

Table 1. Literature comparison across traffic forecasting, flood forecasting, video anomaly detection, RL-based traffic control, and urban knowledge-graph / LLM-agent literature.

2.2 Numbered reference list (matching Table 1)

- [1] Kalyuzhnaya, A., Mityagin, S., Lutsenko, E., et al. “LLM Agents for Smart City Management: Enhancing Decision Support Through Multi-Agent AI Systems.” Smart Cities, 8(1):19, 2025.
- [2] Bouchemoukha, A., et al. “A spatial-temporal graph gated transformer for traffic forecasting.” Transactions on Emerging Telecommunications Technologies, 2024.
- [3] “Traffic congestion predictor: A spatiotemporal graph neural network with congestion-conditional adaptive mechanism and optimization algorithm.” ScienceDirect (Expert Systems with Applications family), 2025.
- [4] “A Sparse Cross Attention-based Graph Convolution Network with Auxiliary Information Awareness for Traffic Flow Prediction.” arXiv preprint, 2023. (Preprint — supporting material only.)
- [5] Transformer-LSTM-Sparrow Search Algorithm urban flood prediction model, applied to Zhengzhou, China. MDPI Water, 2025.
- [6] “Urban flood depth prediction using an improved LSTM model incorporating precipitation forecasting” (Shenzhen case study). Natural Hazards, Springer Nature, 2024.
- [7] Oddo, P.C., Bolten, J.D., Kumar, S.V., Cleary, B. “Deep Convolutional LSTM for improved flash flood prediction.” Frontiers in Water, 6:1346104, 2024.
- [8] “A highly generalizable data-driven model for spatiotemporal urban flood dynamics real-time forecasting based on coupled CNN and ConvLSTM.” Hydrology and Earth System Sciences (HESS), 2026.
- [9] “Video Anomaly Detection in 10 Years: A Survey and Outlook.” arXiv, 2024. (Preprint — supporting material only.)
- [10] Ali, et al. “Real-time video anomaly detection for smart surveillance.” IET Image Processing, 2023.

- [11] Hu, et al. “A multi-agent deep reinforcement learning approach for traffic signal coordination.” IET Intelligent Transport Systems, 2024.
- [12] Cai, C., Wei, M. “Adaptive urban traffic signal control based on enhanced deep reinforcement learning.” Scientific Reports, 14:14116, 2024.
- [13] “Knowledge Graphs for Integrated Urban Data Management in Smart Cities: A Framework for Semantic Interoperability Across Urban Domains” (UrbanKG). Urban Science (MDPI), 2026.
- [14] Ning, Y., Liu, H., Wang, H., Zeng, Z., Xiong, H. “UUKG: Unified Urban Knowledge Graph Dataset for Urban Spatiotemporal Prediction.” NeurIPS, 2024.
- [15] “Fostering Urban Resilience and Accessibility in Cities: A Dynamic Knowledge Graph Approach” (Pirmasens case study). Preprint submitted to Sustainable Cities and Society, 2024. (Preprint — supporting material only.)
- [16] Elnour, M., Ahmad, A.M., Abdelkarim, S., Fadli, F., Naji, K. “Empowering smart cities with digital twins of buildings.” Building Services Engineering Research and Technology, 45(4), 2024.
- [17] “Smart City Digital Twins: A Modular and Adaptive Architecture for Real-Time Data-Driven Urban Management” (Issy-les-Moulineaux case study). IEEE Conference Publication, 2025.
- [18] Ning, Y., Liu, H. “UrbanKGent: A Unified Large Language Model Agent Framework for Urban Knowledge Graph Construction.” NeurIPS, 2024.

2.3 Research gaps

- Modality isolation: traffic-forecasting GNNs ([2],[3],[4]), flood-forecasting LSTM/Transformer hybrids ([5]–[8]), and video anomaly detectors ([9],[10]) are each optimized and evaluated in isolation; none of the surveyed works fuse vision, time-series, and knowledge-graph context in one decision loop.
- Missing-modality robustness is largely untested: forecasting and detection papers assume complete sensor availability at inference time, whereas real municipal deployments routinely lose individual cameras or IoT feeds.
- Simulation-only validation: the strongest reinforcement-learning results for traffic-signal control ([11],[12]) are demonstrated purely in SUMO simulation with no field deployment, so real-world transfer gaps (sensor noise, actuation delay) remain unquantified.
- Explainability is superficial in agentic systems: the LLM multi-agent decision-support work [1] reports strong routing/RAG accuracy but evaluates response quality with G-Eval rather than testing whether explanations change administrator decisions or trust.
- Knowledge-graph integration with predictive models is nascent: KG frameworks ([13],[14],[15],[18]) demonstrate semantic integration and cross-domain querying but stop short of feeding graph context into a live forecasting/RL decision pipeline — exactly the gap this project's fusion-and-orchestration architecture targets.

3. Problem Statement

Given a continuous, multimodal urban data stream — CCTV video (variable frame rate, up to 1080p), IoT sensor time series (traffic, weather, water-level; sampled every 5 minutes across a road/drainage network of a few hundred nodes), GPS/fleet trajectories, and a static urban knowledge graph of infrastructure entities and relations, where one or more modalities (e.g., a subset of cameras or sensor channels) may be missing or degraded at inference time, the system must (a) forecast short-horizon traffic congestion and flood-depth risk per network node/zone, (b) detect and localize visual incidents/anomalies from available video, and (c) output a ranked, explainable set of administrator recommendations (signal re-timing, rerouting, or alert issuance) with an associated confidence score, such that the fused system's forecasting error degrades by less than 10% relative Dice/F1 or by less than 15% relative MAE/RMSE when up to 30% of input modalities are unavailable, benchmarked against single-modality baselines (Graph WaveNet-class traffic GNN, LSTM-class flood model, and a CNN-autoencoder video-anomaly baseline) evaluated on a held-out multi-week window of a pilot city corridor.

This statement is deliberately modeled on the course's "acceptable" example: it names the input modalities and their dimensions/volume, the three output types produced, the specific constraint that motivates the architecture (missing-modality robustness under real deployment conditions), and a quantitative success criterion measured against named single-modality baselines.

4. Proposed System Architecture

4.1 End-to-end pipeline

Figure 1 shows the full pipeline from raw city data to administrator-facing output. Six stages are used: (1) data acquisition across five source types; (2) preprocessing and multimodal fusion, including geo-referencing and missing-data imputation; (3) the cognitive multi-agent model core, comprising a computer-vision incident/anomaly module, a time-series forecasting module, an LLM reasoning/explanation layer that queries the urban knowledge graph via retrieval-augmented generation, and a multi-agent reinforcement-learning simulation/control module, all coordinated by an agent orchestrator; (4) post-processing (scenario simulation, uncertainty scoring, explainability); (5) evaluation against baselines and ablations; and (6) an explainable dashboard/alert interface for city administrators.

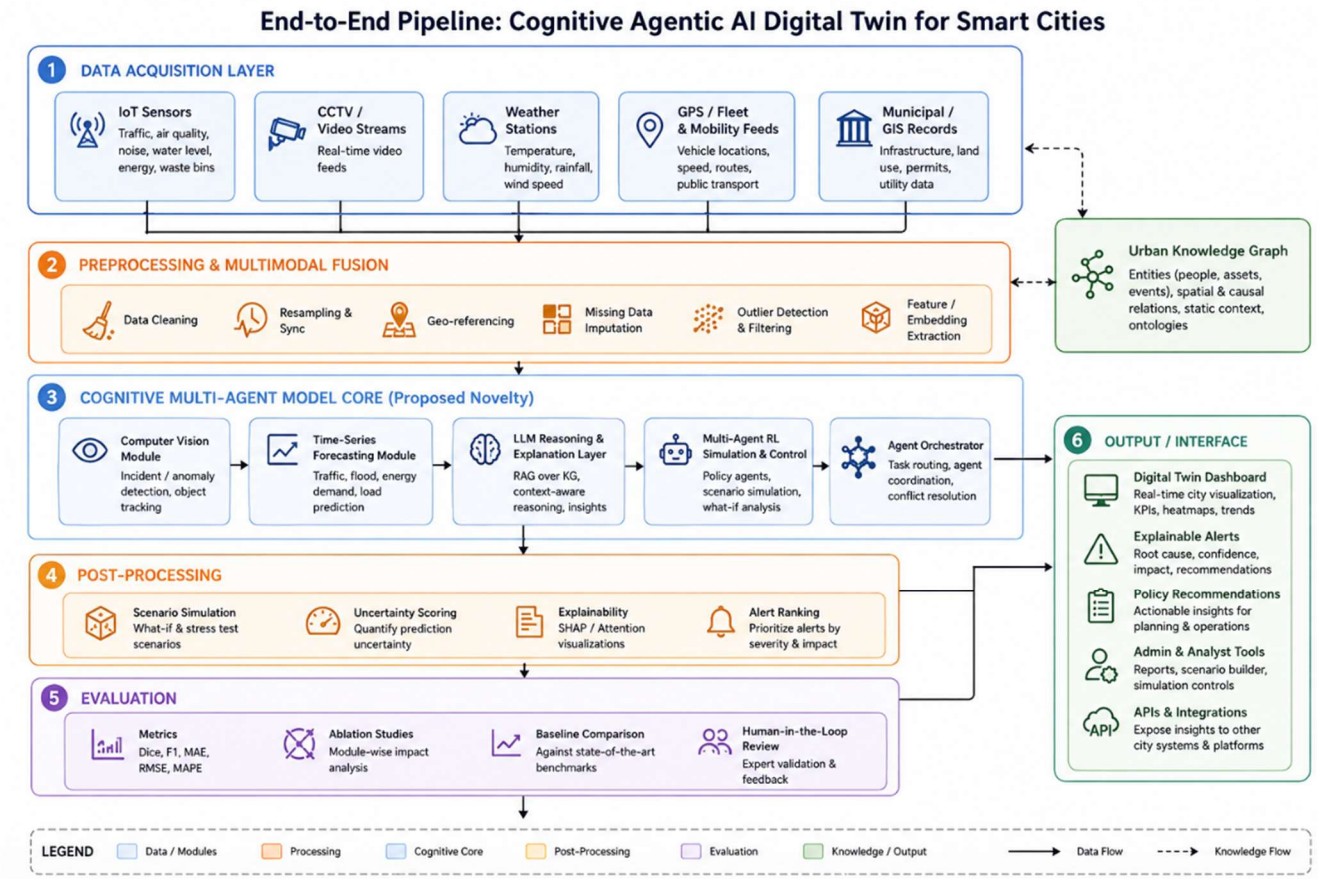


Figure 1. End-to-end pipeline block diagram (data acquisition → preprocessing → model → post-processing → evaluation → output/interface).

4.2 Detailed model diagram

Figure 2 details the model internals with layer names and tensor shapes (B = batch size). Stage A independently encodes each modality: video clips via a 3D-CNN/ViT-B backbone, IoT time series via a temporal graph neural network in the Graph WaveNet family, GPS trajectories via a GRU encoder, the urban-knowledge-graph subgraph via a relational GNN (R-GCN), and free-text reports via a frozen LLM text encoder. Stage B is the proposed novelty: a cross-attention fusion transformer with gated modality-dropout training, which is trained to tolerate missing modalities by randomly zeroing modality embeddings during training and learning attention gates that reweight the remaining modalities — directly targeting the missing-modality constraint in the problem statement. Stage C routes the fused embedding to four specialized agents (forecasting, detection, simulation/RL, and LLM-based reasoning), coordinated by an orchestrator that performs conflict resolution and confidence weighting. Stage D produces a ranked recommendation vector with an explanation, surfaced through a human-in-the-loop dashboard.

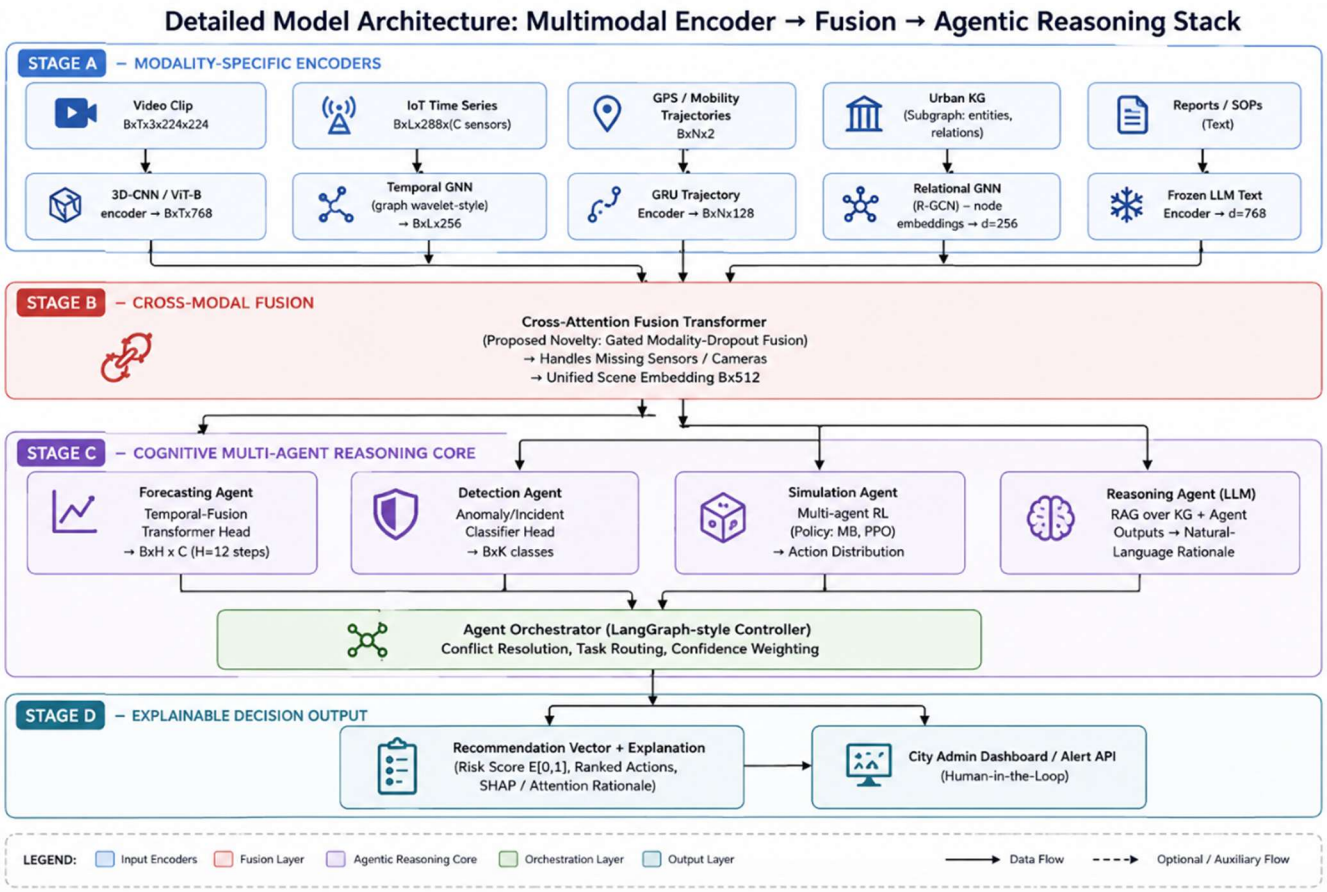


Figure 2. Detailed model architecture with per-stage tensor shapes; the cross-attention fusion transformer with gated modality-dropout (Stage B, outlined in red) is the proposed point of novelty.

4.3 Design justification

- Graph-based temporal encoder for IoT/traffic data: graph convolutional approaches consistently outperform grid-based CNN/RNN traffic models because they respect road-network topology rather than distorting it into a regular grid, as shown across the GNN traffic-forecasting literature surveyed in §2 ([2],[3],[4]).
- Transformer-LSTM hybrid for flood forecasting: hybrid architectures that pair a Transformer's global feature extraction with an LSTM's temporal modeling report NSE > 0.94 on real waterlogging data ([5]), motivating a comparable hybrid rather than a single-architecture flood head.
- Cross-attention fusion with modality dropout: none of the surveyed single-modality models are evaluated under sensor/camera loss, which the research gap in §2.3 identifies as unaddressed; gated modality-dropout fusion is a direct, testable hypothesis for closing that gap rather than an unmotivated addition.
- Multi-agent RL for signal/simulation control: multi-agent DDQN and enhanced dueling-DQN variants report 40–60% reductions in waiting time and 10–18% speed increases in SUMO ([11],[12]); using an RL-based simulation agent, rather than a fixed heuristic controller, follows this evidence while acknowledging (per the same papers) that gains are currently simulation-only.

- LLM reasoning layer with RAG over the knowledge graph: the closest prior system to this project's decision-support goal ([1]) shows RAG grounding improves response accuracy by 17–55% over an ungrounded LLM baseline, justifying retrieval over a structured urban KG rather than relying on the LLM's parametric knowledge alone.

4.4 Planned experimental setup

- Datasets: a public traffic-forecasting benchmark (e.g., PeMS-Bay/METR-LA class) for the forecasting agent; a public or simulated urban-flood dataset (rainfall-inundation pairs, following the Zhengzhou/Shenzhen study designs in [5],[6]) for the flood head; UCF-Crime/ShanghaiTech for the video-anomaly detection agent; a SUMO network (e.g., a 4×4 grid, following [11],[12]) for the RL simulation agent; and a constructed small-scale urban knowledge graph (following the UUKG schema [14]) for the reasoning agent.
- Split strategy: chronological train/validation/test splits for all time-series and video data (no shuffling across time) to avoid leakage; held-out corridor/zone splits to test spatial generalization.
- Evaluation metrics: MAE/RMSE for forecasting heads; F1/AUC for the video-anomaly detector; average waiting time, queue length, and throughput for the RL agent; G-Eval-style response-quality scoring for the LLM reasoning layer, following [1].
- Baselines: Graph WaveNet-class GNN (traffic), single-branch LSTM (flood), CNN-autoencoder (video anomaly), fixed-time and single-agent DQN signal control (RL), and an ungrounded LLM without RAG (reasoning layer).
- Hardware: single workstation-class GPU (≥16 GB VRAM) for module-level training; cloud GPU credits (student tier) for full multi-agent fine-tuning runs, scoped to the feasibility budget in §5.
- Ablation studies planned: (i) fusion transformer with vs. without modality dropout, evaluated under synthetic modality-masking; (ii) orchestrator with vs. without confidence weighting; (iii) reasoning agent with vs. without RAG over the knowledge graph; (iv) RL simulation agent vs. fixed-time signal baseline on the same SUMO network.

5. Feasibility Note

Compute available: a single CUDA GPU with ≥16 GB VRAM is sufficient for training each modality-specific encoder and the fusion transformer at the batch sizes shown in Figure 2; full multi-agent RL training (SUMO rollouts) is CPU-bound and can run in parallel on standard lab machines, with cloud GPU credits reserved for final fine-tuning of the LLM-reasoning and fusion stages.

Dataset size and access status: traffic-forecasting benchmarks (PeMS-Bay/METR-LA class) and video-anomaly benchmarks (UCF-Crime, ShanghaiTech) are publicly downloadable today. Flood data at the resolution used in [5]/[6] (per-district waterlogging depth) is not publicly available for our target city and will initially be substituted with a public rainfall-inundation dataset or a physics-simulator-generated dataset, consistent with the surrogate-modeling approach in [8]. The urban knowledge graph will be built from public GIS/OpenStreetMap layers, following the UUKG construction approach [14].

Estimated training time: modality encoders and the fusion transformer, 6–12 GPU-hours each on the target GPU class; RL simulation agent, 12–24 CPU-hours for a 4×4 SUMO grid to reach the convergence behavior reported in [11]/[12]; total estimated wall-clock budget across DA1–DA3 milestones is under one week of non-continuous compute.

Identified risks and fallback plan:

- Real flood data access risk — fallback to simulator-generated inundation data as in [8];
- RL sim-to-real gap — restrict claims to simulation results and report this limitation explicitly, consistent with the same caveat noted for [11],[12];
- LLM hallucination in the reasoning layer — mitigate via RAG grounding over the knowledge graph (per [1]) and require the orchestrator to attach a confidence score rather than an unqualified recommendation;
- Missing-modality evaluation requires synthetic masking since real deployments with controlled sensor outages are unavailable — fallback to masking test data programmatically rather than collecting live outage data.

Contribution matrix

- AKSHARA SV 24BRS1341 : [literature survey + problem statement]
- SWETHA DIVI 24BRS1334: [architecture diagrams + design justification]
- RAHUL VISWANATHAN 24BRS1101: [feasibility note + dataset access + repo setup]